Script-Controlled ACL - Restrict Record Access Based on Field Value

1:Creation of Users and Roles

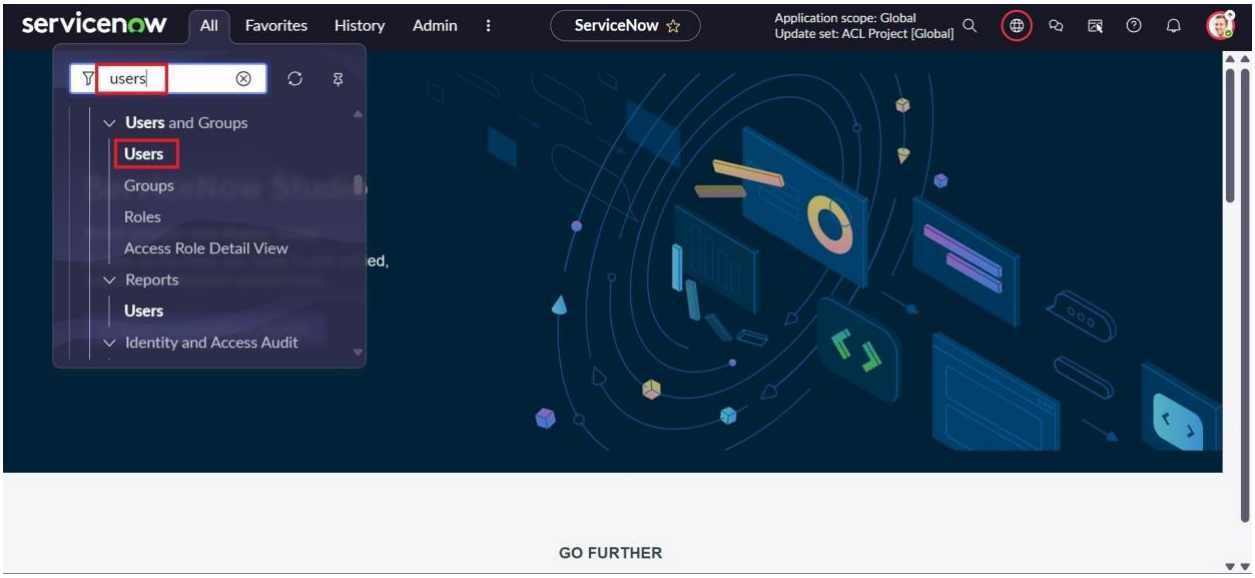
Description

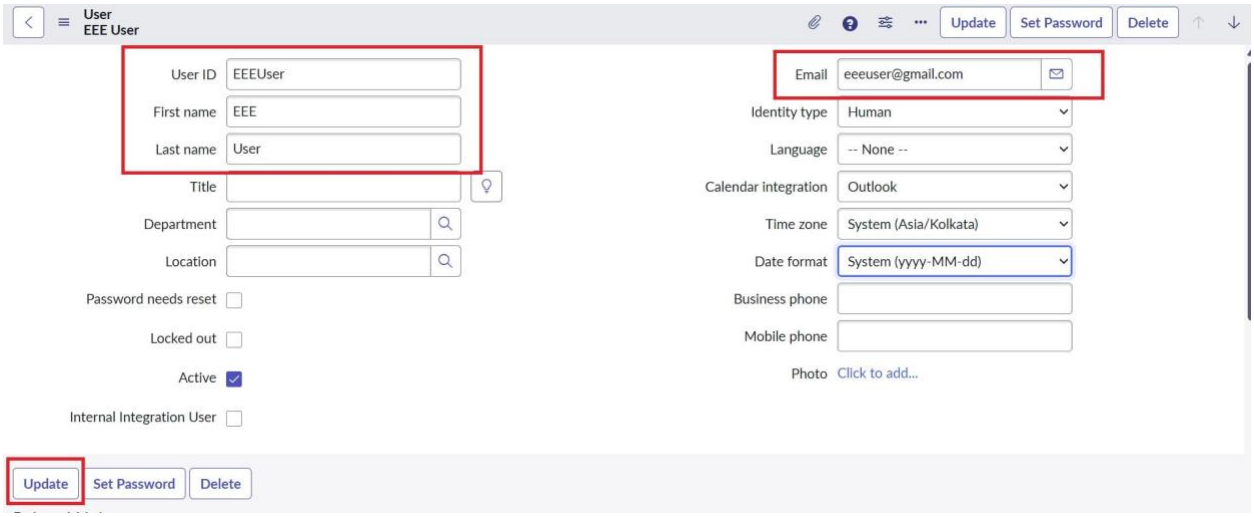
In this milestone we have to create Users and Roles as per the requirement

Users Creation

Steps

- **Log in** to the ServiceNow instance with **admin access**.
- **Create a test user:**
- Navigate to **User Administration → Users**
- Click **New**
- **User ID**: EEE User
- **First name**: EEE
- **Last name**: User
- **Email**: eeeuser@gmail.com
- Save the record

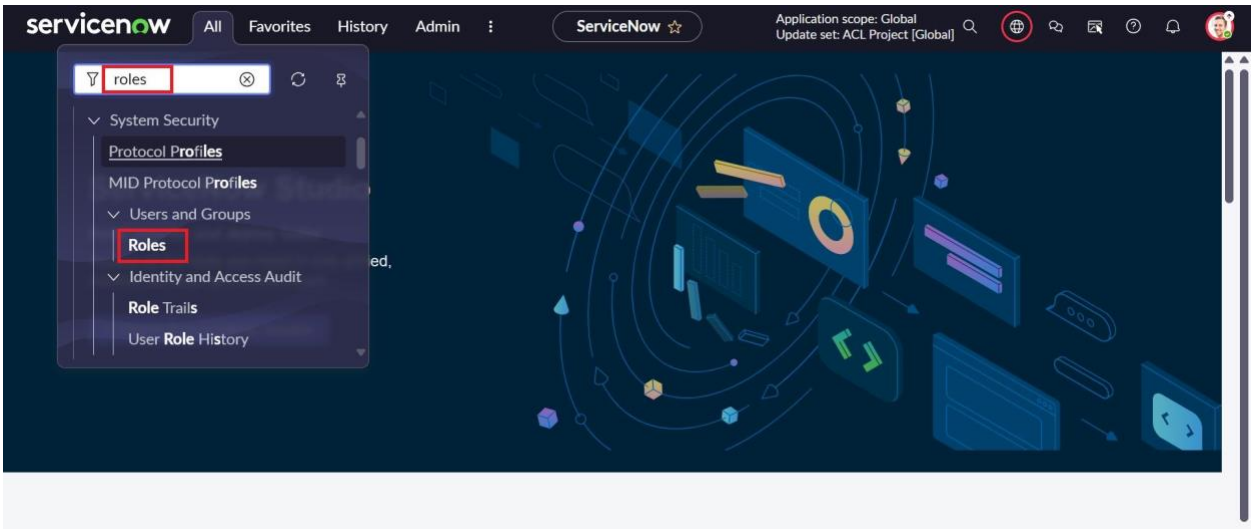

servicenow
All
Favorites
History
Admin
ServiceNow
Application scope: Global
Update set: ACL Project [Global]
users
Users and Groups
Users
Groups
Roles
Access Role Detail View
Reports
Users
Identity and Access Audit
GO FURTHER

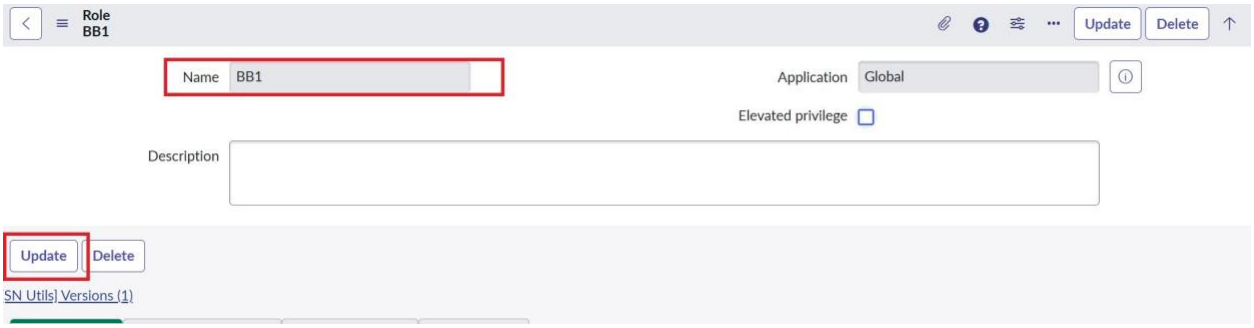

User
EEE User
Update
Set Password
Delete
User ID
EEEUser
First name
EEE
Last name
User
Title
Department
Location
Password needs reset
Locked out
Active
Internal Integration User
Email
eeeuser@gmail.com
Identity type
Human
Language
-- None --
Calendar integration
Outlook
Time zone
System (Asia/Kolkata)
Date format
System (yyyy-MM-dd)
Business phone
Mobile phone
Photo
Click to add...
Update
Set Password
Delete

Roles Creation

Steps

- Navigate to **User Administration → Roles**
- Click **New**
- **Name**: Enter role name as bb1
- **Description**: Brief role purpose
- Click **Submit**
- **Create other roles also like bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User





2:Tables Creation

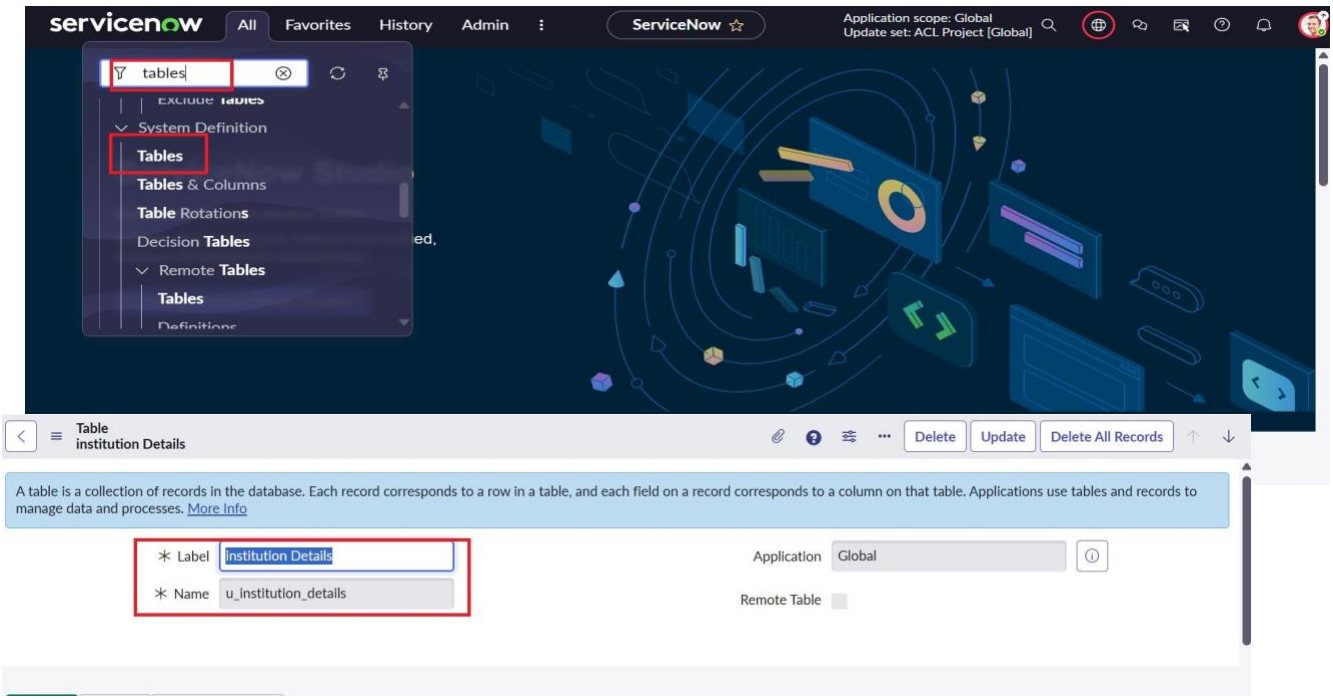
Description

In this phase we have create a custom table to achieve the requirement

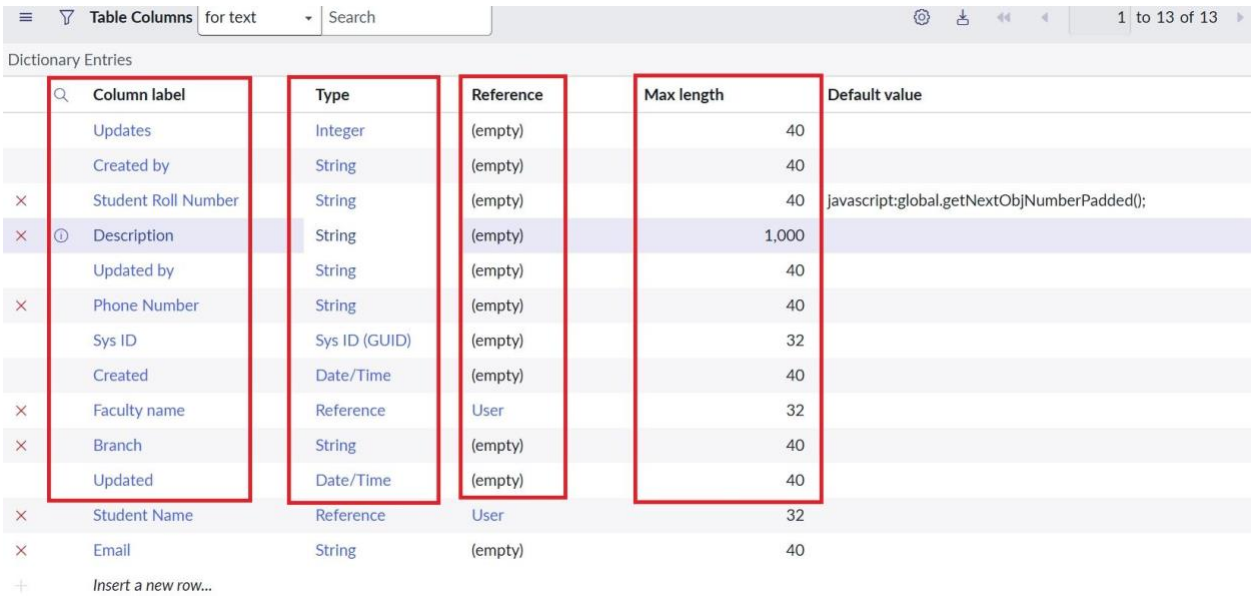
Creation of Table

Steps

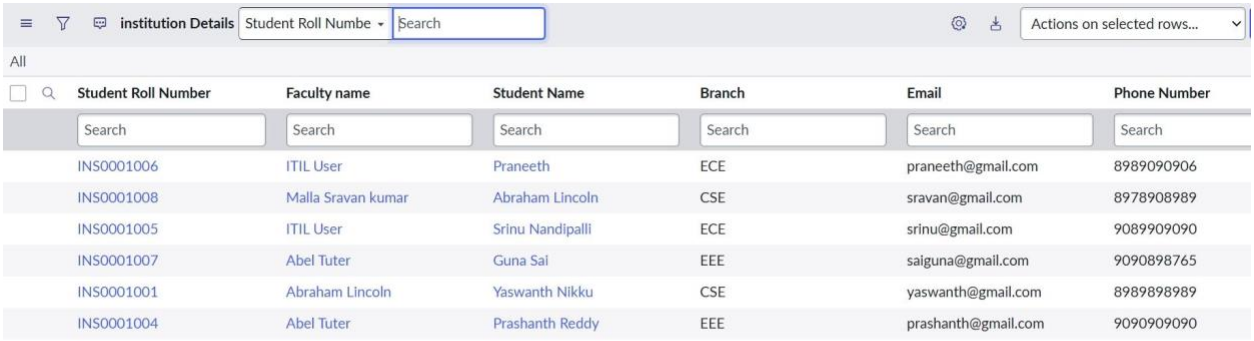
- In the Application Navigator, search for System Definition → Tables.
- Click New and create a custom table with the following details
- Label: Institution Details
- Name: u_institution_details
- Extends for : false
- Save or Submit the form



- In the columns tab, user Insert a New row method to create the mentioned fields as shown below picture
- The following fields to the u_institution_details table
- Student Roll Number– Auto Number
- Student Name – Reference- User
- Faculty Name – Reference- User
- Branch – Choice (ECE, EEE, CSE)
- Email– String
- Phone Number– String
- Description– Multi String
- Save the Table



- Click on the Show List in the Related Links
- Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).


institution Details
Student Roll Numbe
Search
Actions on selected rows...
All
Student Roll Number
Faculty name
Student Name
Branch
Email
Phone Number
Search
INS0001006
ITIL User
Praneeth
ECE
praneeth@gmail.com
8989090906
INS0001008
Malla Sravan kumar
Abraham Lincoln
CSE
sravan@gmail.com
8978908989
INS0001005
ITIL User
Srinu Nandipalli
ECE
srinu@gmail.com
9089909090
INS0001007
Abel Tuter
Guna Sai
EEE
saiguna@gmail.com
9090898765
INS0001001
Abraham Lincoln
Yaswanth Nikku
CSE
yaswanth@gmail.com
8989898989
INS0001004
Abel Tuter
Prashanth Reddy
EEE
prashanth@gmail.com
9090909090

3: Creation of Access Control List-READ

Description

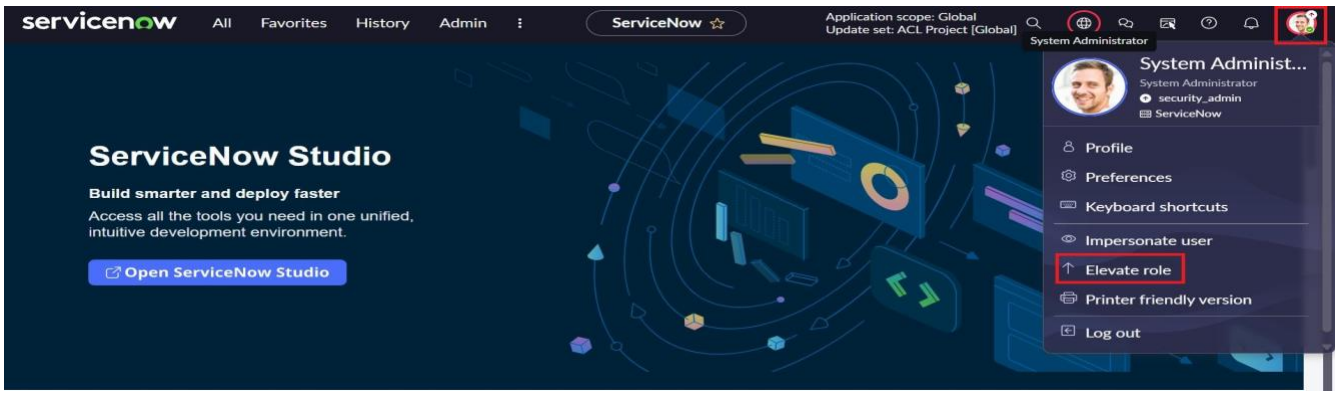
In this phase we have to configure ACL's to restrict the particular table and fields

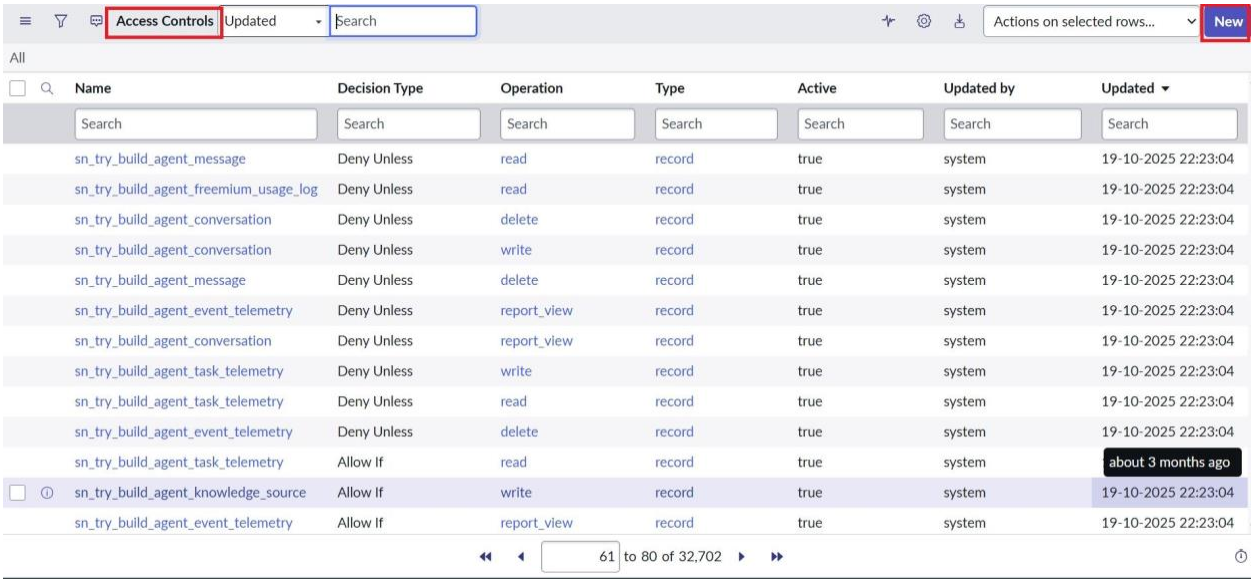
ACL Configuration

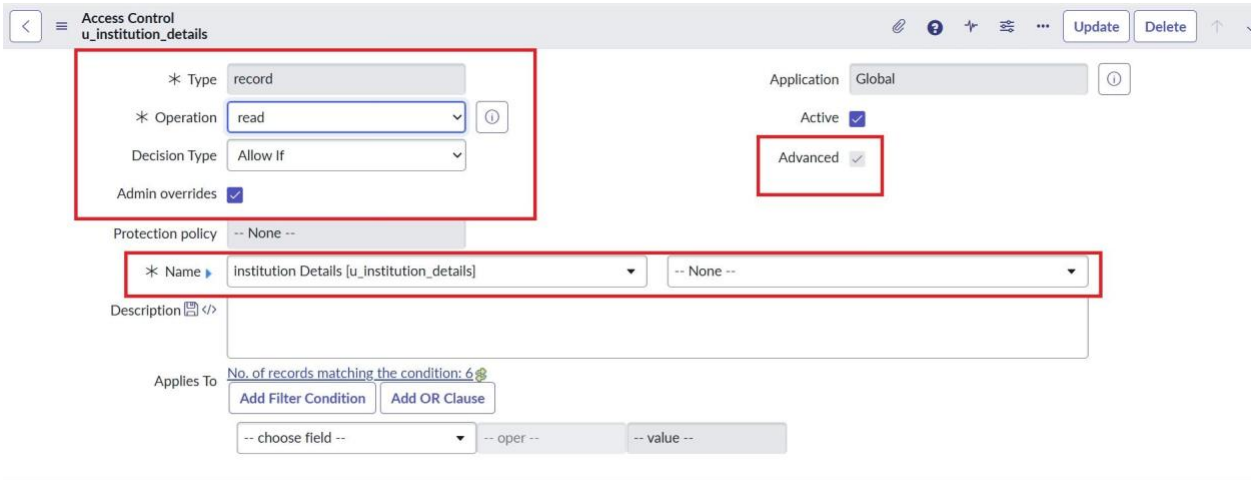
Steps

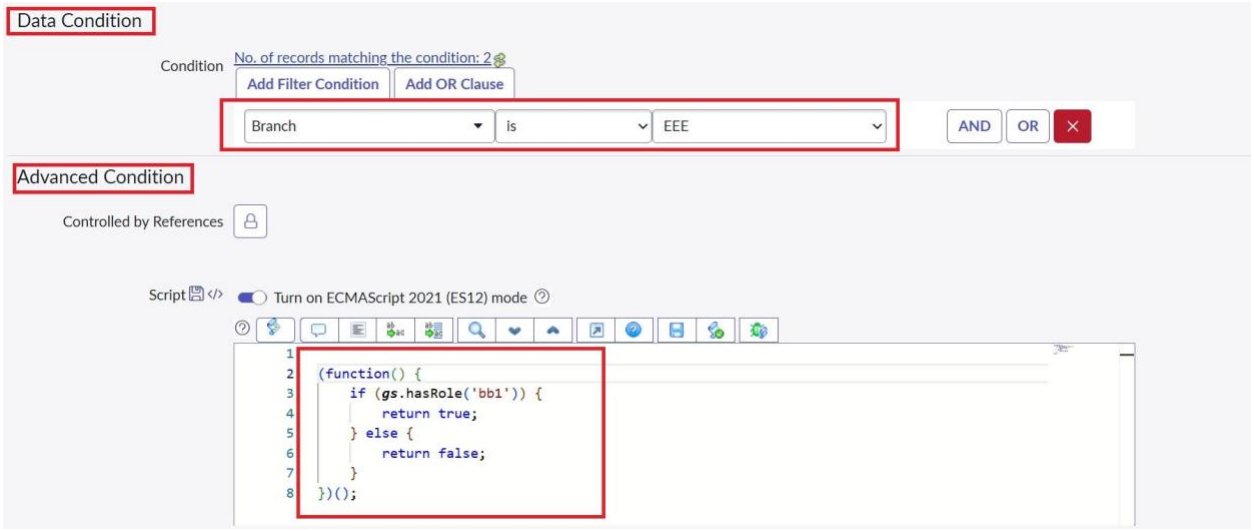
- In the Application Navigator, search for System Security → Access Control (ACL).
- Elevate “Security_admin” role
- Click New to create a new ACL.
- Enter the following details
- Type: record
- Operation: read
- Name: u_institution_details
- Active: true
- Advanced: true
- In the Requires role related list, add: bb1 (custom role)
- Add Data condition as Branch is EEE
- In the Script field, add the following script
- Save the ACL form

```
(function () {
    // Allow admin users full access
    if (gs.hasRole('admin')) {
        return true;
    }
    // Allow only EEE branch users to se
e EEE records
    if (gs.hasRole('bb1')){
        return true;
    }
    // Deny access for all others
    return false;
})();
```



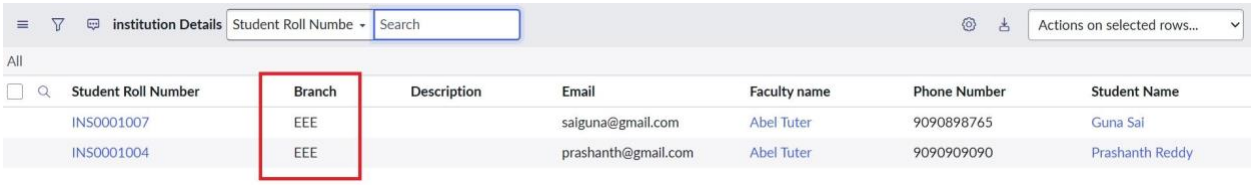

Access Controls
Updated
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
sn_try_build_agent_message Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_freemium_usage_log Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_message Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Allow If read record true system about 3 months ago
sn_try_build_agent_knowledge_source Allow If write record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Allow If report_view record true system 19-10-2025 22:23:04
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Access Control
u_institution_details
Update
Delete
Type record
Operation read
Decision Type Allow If
Admin overrides
Application Global
Active
Advanced
Protection policy -- None --
Name institution Details [u_institution_details]
-- None --
Description
Applies To
No. of records matching the condition: 6
Add Filter Condition
Add OR Clause
-- choose field --
-- oper --
-- value --


Data Condition
Condition
No. of records matching the condition: 2
Add Filter Condition
Add OR Clause
Branch
is
EEE
AND
OR
Advanced Condition
Controlled by References
Script
Turn on ECMAScript 2021 (ES12) mode
(function() {
if (gs.hasRole('bb1')) {
return true;
} else {
return false;
}
})();

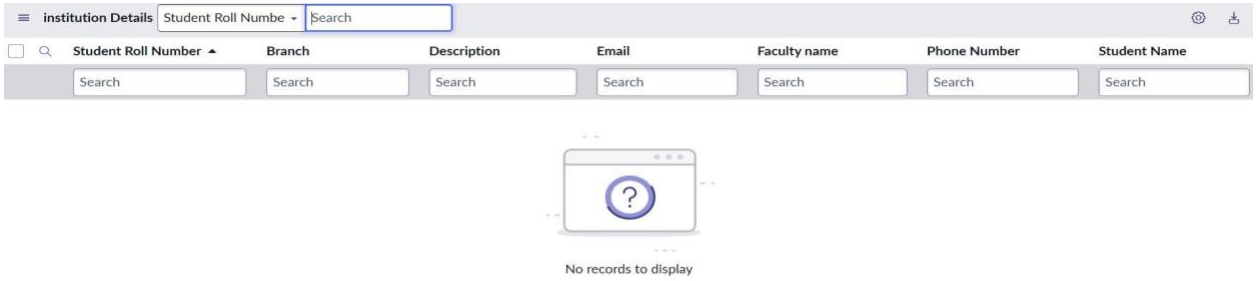
- Impersonate EEE users and open the Student Records list.
- Copy the table name u_institution_details.list/LIST and paste in the All menu.
- Impersonate Admin and open the Student Records list.

Verify : User with bb1 role can view only EEE branch records

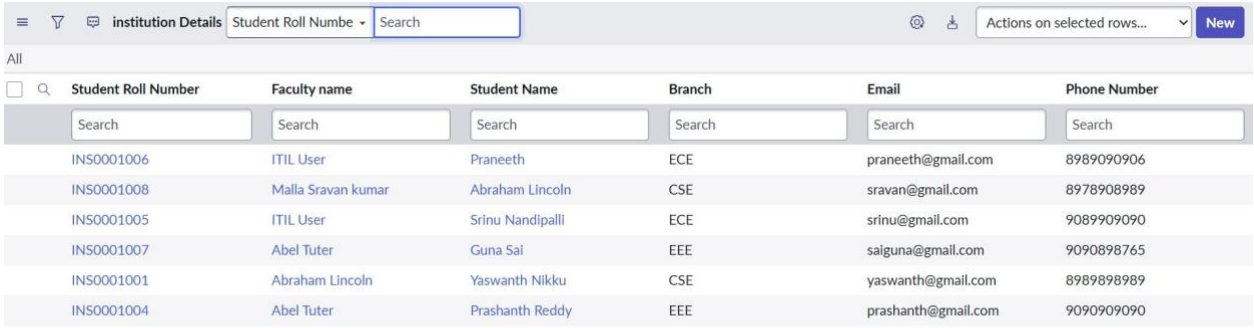


Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE		saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

- User without the role cannot view any records



- Admin user can view **all records regardless of branch**



Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

4: Creation of Access Control List-CREATE

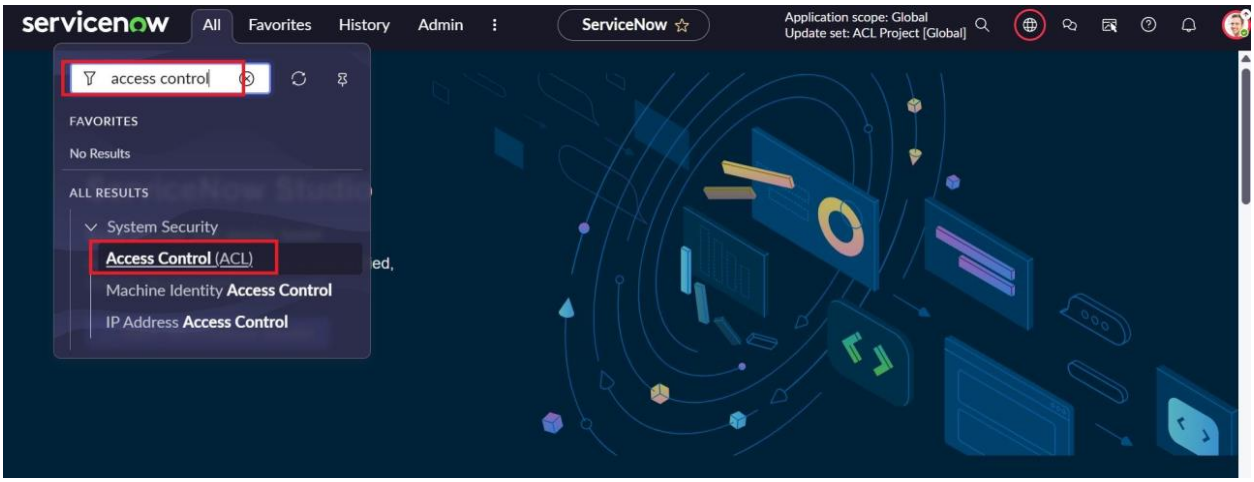
Description

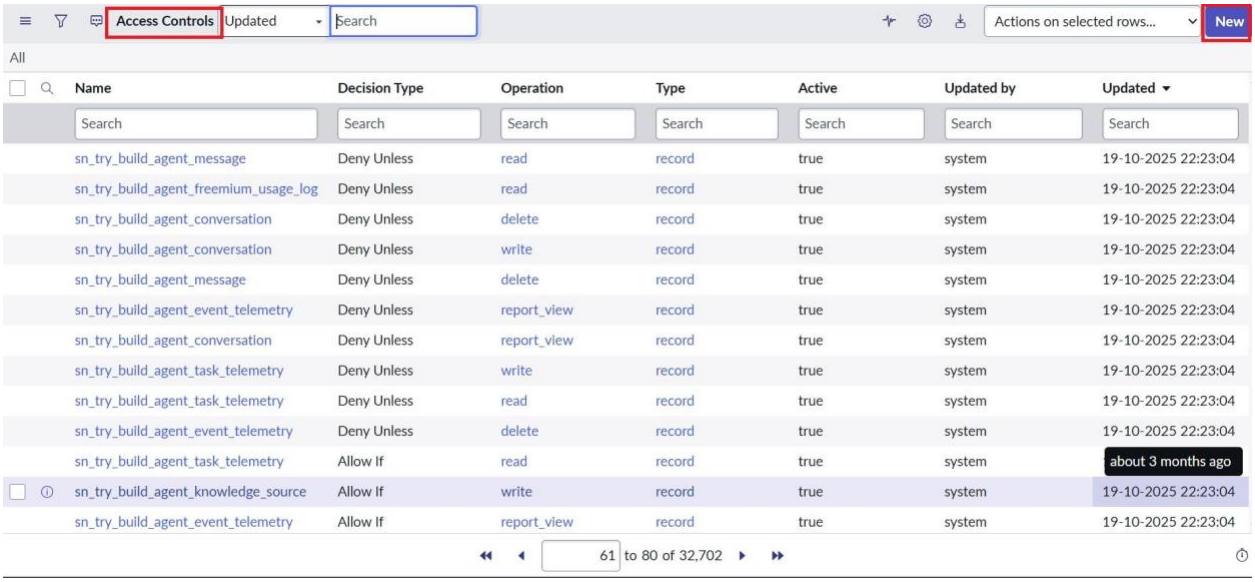
In this phase we have to create Create ACL to acheive the requirement

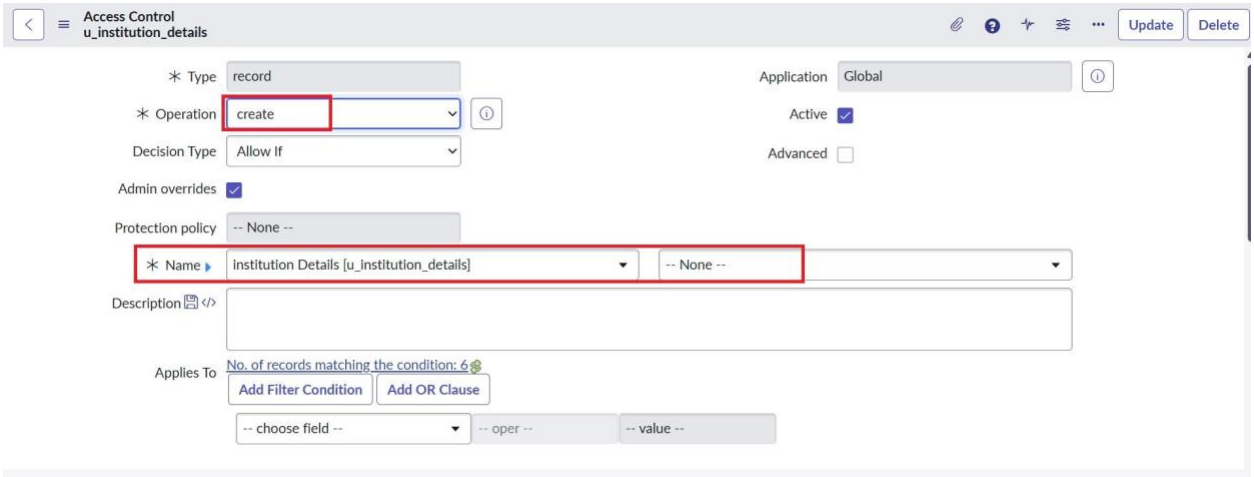
Creation of ACL-Create

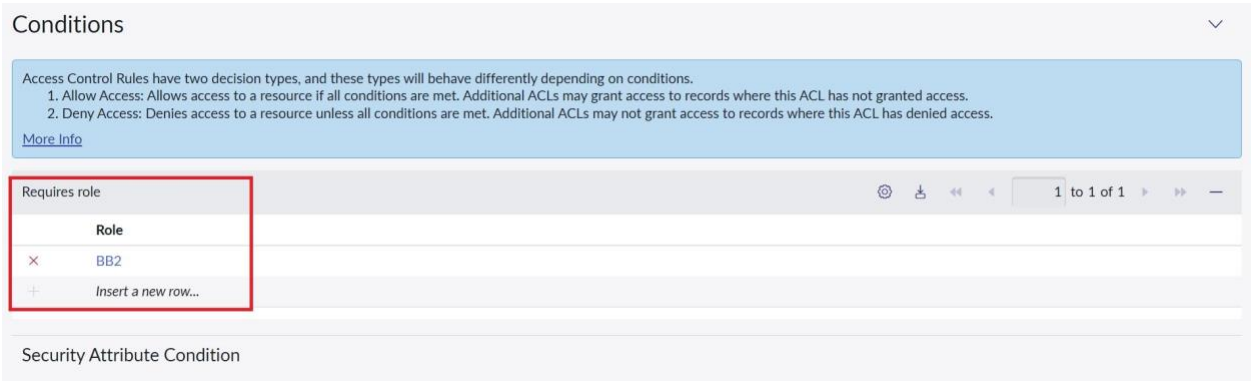
Steps

- Click New to create a new ACL(Create)
- Enter the following details
- Type: record
- Operation: create
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb2 (custom role)
- No need to add any Data condition
- Save the ACL Record



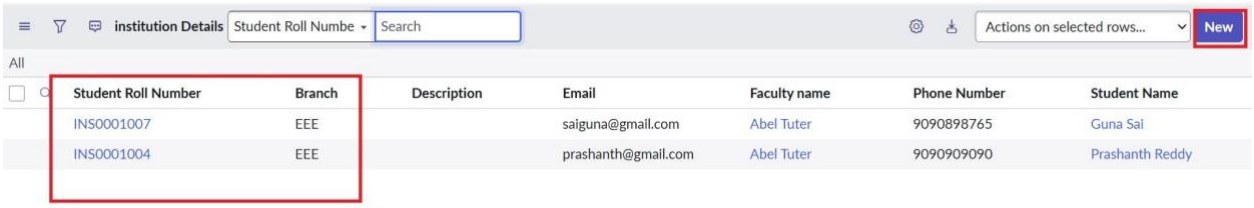

Access Controls
Updated
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
sn_try_build_agent_message Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_freemium_usage_log Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_message Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Allow If read record true system about 3 months ago
sn_try_build_agent_knowledge_source Allow If write record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Allow If report_view record true system 19-10-2025 22:23:04
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Access Control
u_institution_details
Update
Delete
Type record
Application Global
Operation create
Active
Decision Type Allow If
Advanced
Admin overrides
Protection policy -- None --
Name institution Details [u_institution_details]
-- None --
Description
Applies To
No. of records matching the condition: 6
Add Filter Condition
Add OR Clause
-- choose field --
-- oper --
-- value --


Conditions
Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
More Info
Requires role
1 to 1 of 1
Role
BB2
Insert a new row...
Security Attribute Condition

Verify :

- User with **bb1 &bb2 roles** can view **only EEE branch records & New** Button



5: Creation of Access Control List-WRITE

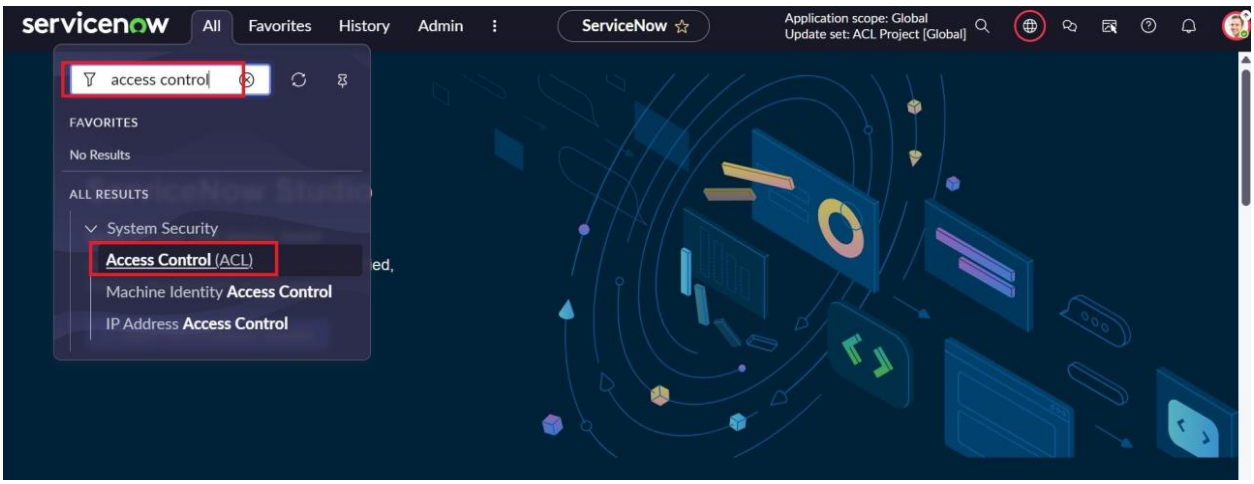
Description

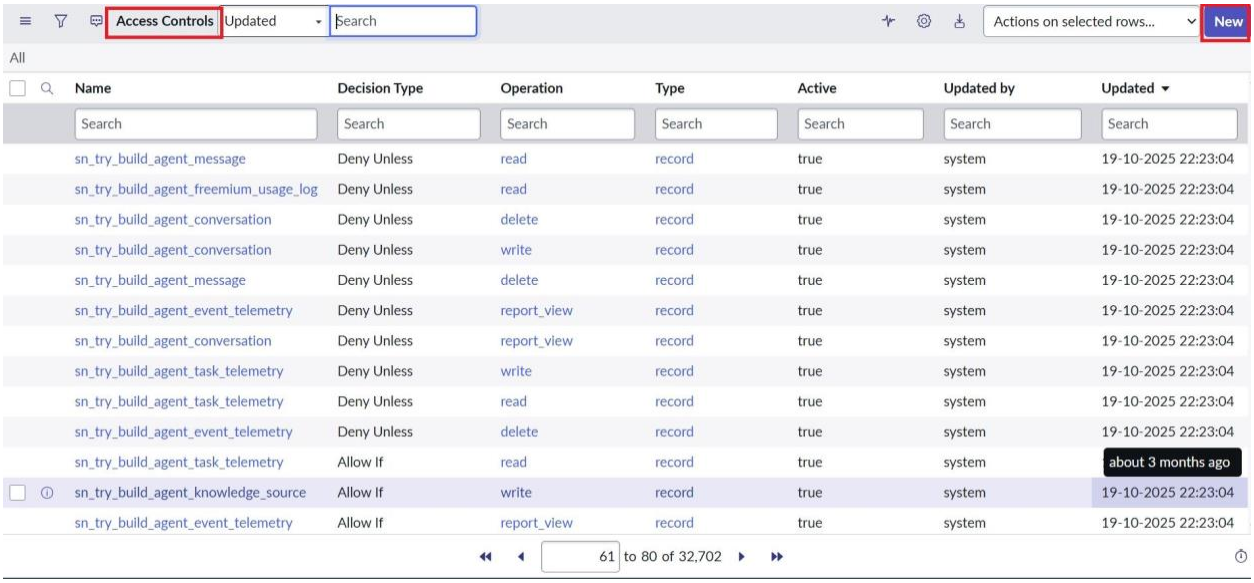
In this phase we have to create a Write ACL to achieve the requirement

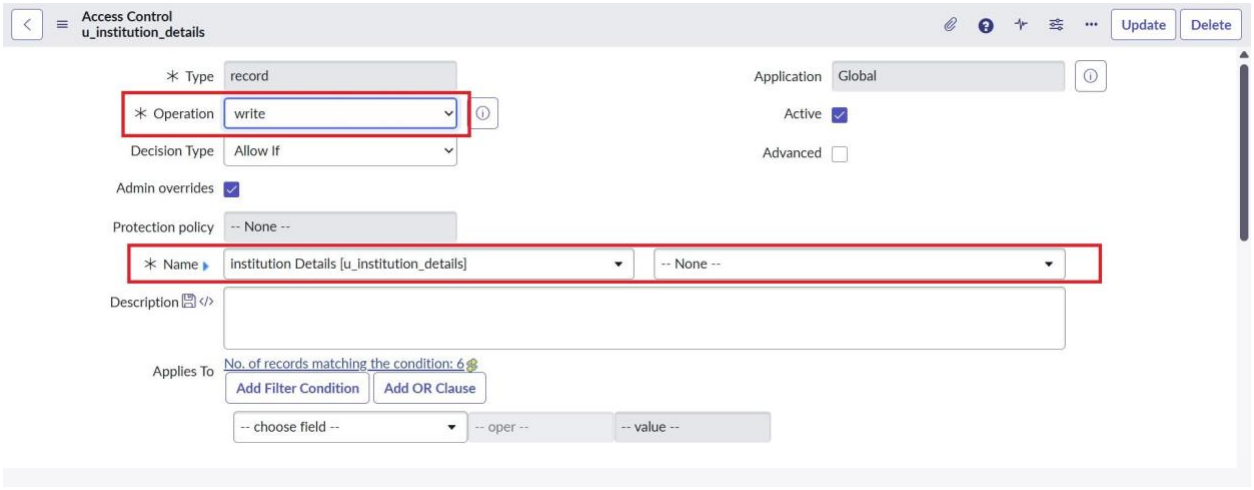
Creation of ACL-Write

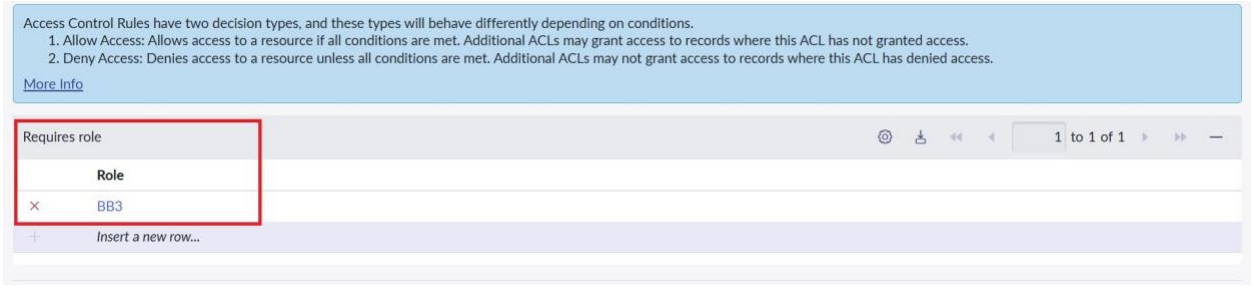
Steps

- Click New to create a new ACL(Write)
- Enter the following details
- Type: record
- Operation: Write
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb3 (custom role)
- No need to add any Data condition
- Save the ACL Record



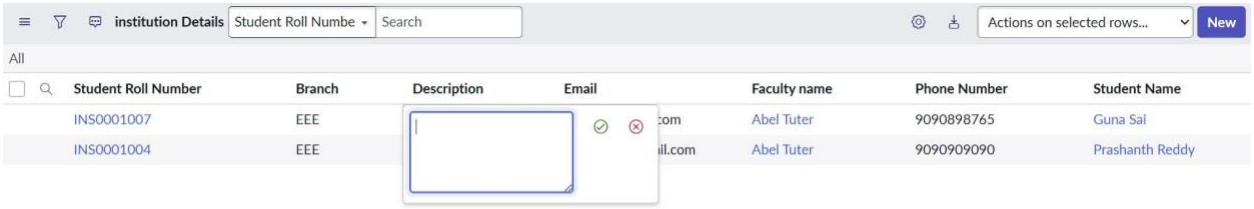

Access Controls
Updated
Search
Actions on selected rows...
New
All
Name
Decision Type
Operation
Type
Active
Updated by
Updated
sn_try_build_agent_message Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_freemium_usage_log Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_message Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_conversation Deny Unless report_view record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless write record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Deny Unless read record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Deny Unless delete record true system 19-10-2025 22:23:04
sn_try_build_agent_task_telemetry Allow If read record true system about 3 months ago
sn_try_build_agent_knowledge_source Allow If write record true system 19-10-2025 22:23:04
sn_try_build_agent_event_telemetry Allow If report_view record true system 19-10-2025 22:23:04
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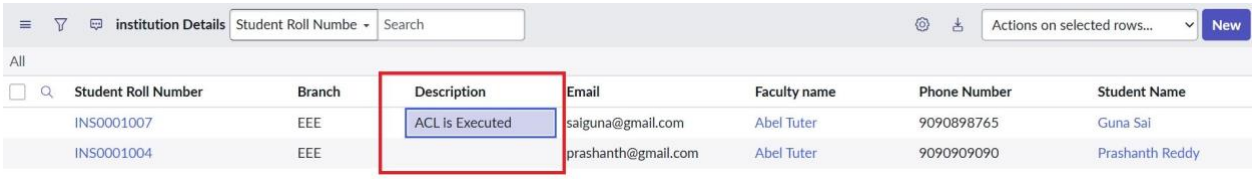

Access Control
u_institution_details
Update
Delete
Type record
Application Global
Operation write
Active
Decision Type Allow If
Advanced
Admin overrides
Protection policy -- None --
Name institution Details [u_institution_details]
-- None --
Description
Applies To
No. of records matching the condition: 6
Add Filter Condition
Add OR Clause
-- choose field --
-- oper --
-- value --


Access Control Rules have two decision types, and these types will behave differently depending on conditions.
1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
More Info
Requires role
1 to 1 of 1
Role
BB3
Insert a new row...

Verify :

- User with **bb1 &bb2 &bb3 roles** can view and edit the **EEE branch records along with New** Button





6: Creation of Access Control List-DELETE

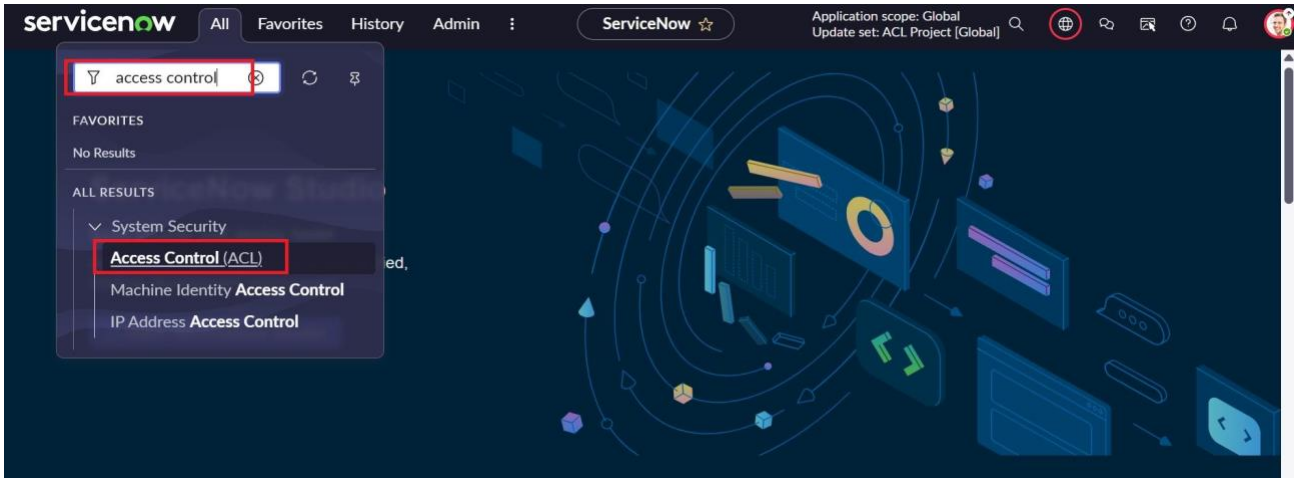
Description

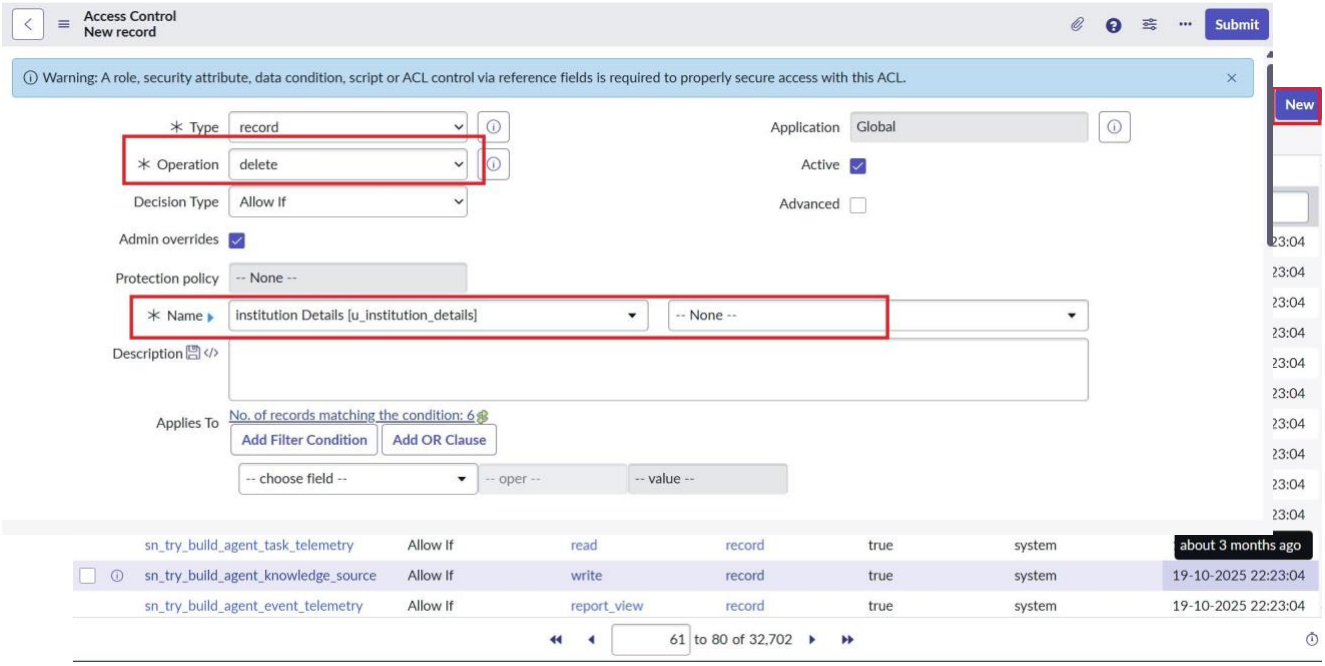
In this phase we have to create delete ACL to achieve the requirement

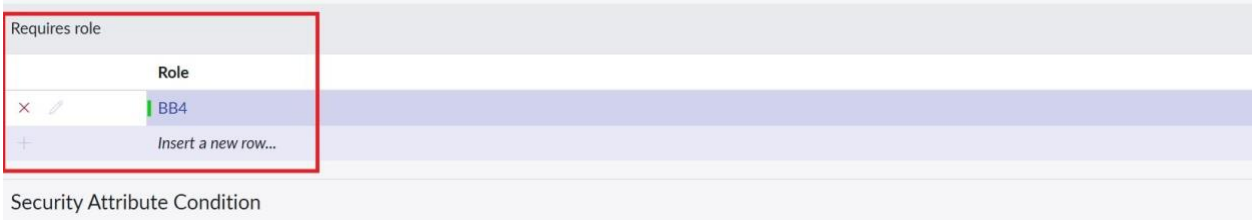
Creation of ACL-DELETE

Steps

- Click New to create a new ACL(Delete)
- Enter the following details
- Type: record
- Operation: Delete
- Name: u_institution_details
- Active: true
- In the Requires role related list, add: bb4 (custom role)
- No need to add any Data condition
- Save the ACL Record

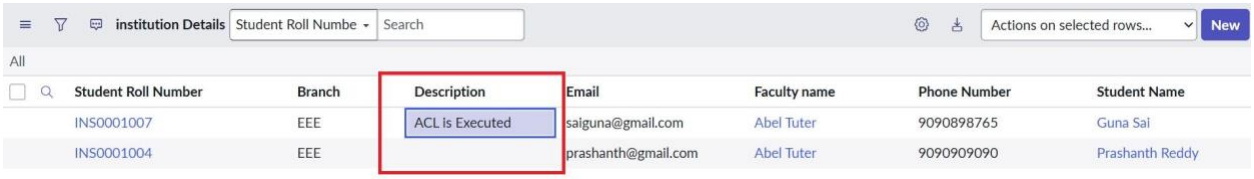

servicenow
All
Favorites
History
Admin
ServiceNow
Application scope: Global
Update set: ACL Project [Global]
access control
FAVORITES
No Results
ALL RESULTS
System Security
Access Control (ACL)
Machine Identity Access Control
IP Address Access Control

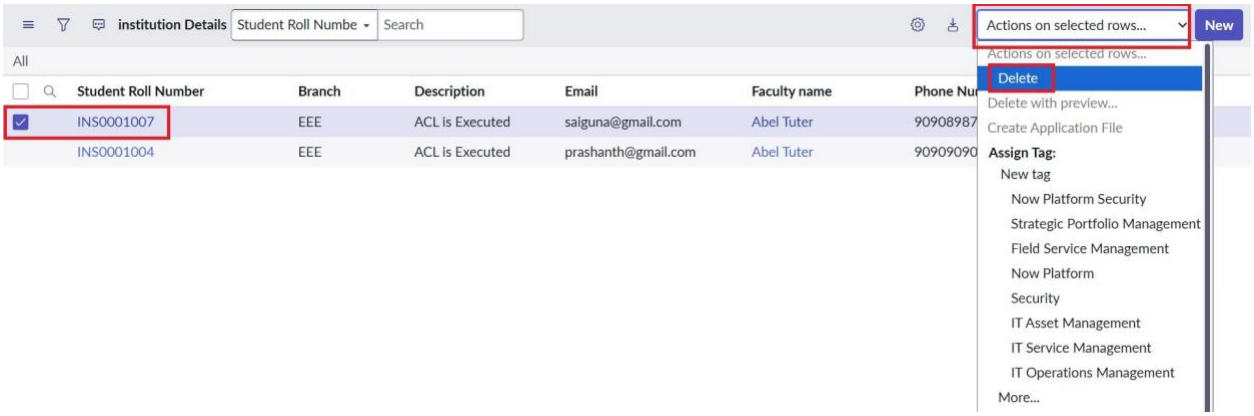

Access Control
New record
Submit
Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.
Type
record
Application
Global
Operation
delete
Active
Decision Type
Allow If
Advanced
Admin overrides
Protection policy
-- None --
Name
institution Details [u_institution_details]
-- None --
Description
Applies To
No. of records matching the condition: 6
Add Filter Condition
Add OR Clause
-- choose field --
-- oper --
-- value --
New
sn_try_build_agent_task_telemetry
Allow If
read
record
true
system
about 3 months ago
sn_try_build_agent_knowledge_source
Allow If
write
record
true
system
19-10-2025 22:23:04
sn_try_build_agent_event_telemetry
Allow If
report_view
record
true
system
19-10-2025 22:23:04
61 to 80 of 32,702

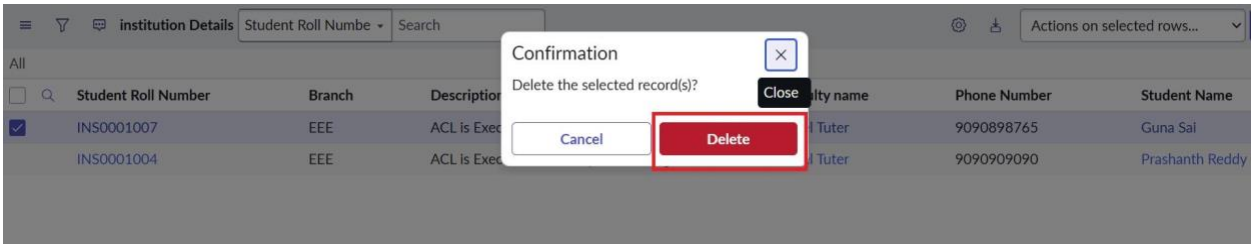

Requires role
Role
BB4
Insert a new row...
Security Attribute Condition

Verify :

- Users with **bb1 &bb2 &bb3&bb4 roles** can see, create and edit along with delete the **EEE branch records.**







7: Conclusion

Description

In this phase we have to conclude the project

Project Outcome

Description

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces